ArticleMiner: Supplementary Appendix

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A Output Schemas

The ArticleMiner GeoChem ground-truth schema contains 209 columns: 63 metadata/provenance fields and 146 element columns (73 elements × {ppm value, detection limit}). The pipeline's output Excel allows extra optional fields (uncertainty, method, alternative units), but evaluation pulls only the named-field projection (§3), so additional pipeline-emitted columns are ignored rather than penalized — pipeline and baseline are compared head-to-head on the same 209-column projection. The other three benchmarks each define a tuple shape: ChemTables uses ⟨*cell_index, value, type, target, treatment, unit*⟩ with *type* ∈ {IC_{50},EC_{50},GI_{50},MIC}; DiSCoMaT uses ⟨*sample_id, component, percentage, unit*⟩ with the unit in {mol%, wt%}; MLTables uses an *entry-type*-tagged tuple in {Result,Data Stat.,Hyper-parameter,Other} with type-dependent attribute fields (e.g. Result carries *model, dataset, metric, task*).

B Per-Domain External-Benchmark Detail

DiSCoMaT (materials science, n=111). ArticleMiner reaches 81.0 F_1 with Opus 4.6 (+21.5 over the same-LLM PDF baseline) and 79.3 with Sonnet 4.6 (+14.8); precision is 97–98% on both backbones because the materials-science ontology rejects any tuple whose component is not a recognized oxide and whose unit is not in {mol%, wt%}. Even on raw-PDF input ArticleMiner exceeds the published Gupta GNN baseline on pre-parsed tables (70.04 F_1), with p<0.001 (Table 4). The recall ceiling is parsing, not semantics: median per-paper F_1 is 96.7, and the mean is dragged down by 32 of 111 papers returning zero predictions because docling fails on 1999–2015 Elsevier PDF layouts, and the environment lacks the YOLO weights mineru would need to fall back to OCR. With those 32 excluded, the remaining 79 papers reach 91.8 F_1; the pre-parsed Sonnet ArticleMiner score of 91.3 (Table 1) corroborates the diagnosis. Open 7–8B LLMs trail substantially under the same pipeline: Qwen-2.5-7B 57.8 F_1 (−21.5 vs. Sonnet), Mistral-7B-v0.2 51.9 (−27.4), and Llama-3.1-8B 18.2 (−61.1); Llama collapses on recall (R=10.3%) yet still hits P=79.0% via the ontology gate (Fig. 6).

MLTables (machine learning, n=15). A far-transfer test: ML papers report model×dataset matrices with prose density that lets the few-shot baseline recover a non-trivial fraction of the gold without table understanding. ArticleMiner still beats the same-LLM PDF baseline with paired Δ=+14.4 F_1 (Opus 55.9 vs. Opus few-shot 41.6; per-paper bootstrap, p=0.074 at n=15; Table 4). The smaller margin matches the cross-domain hypothesis: architectural value scales inversely with the share of gold recoverable from the inline narrative. Open 7–8B LLMs nearly match Sonnet here: Qwen-2.5-7B 50.2 F_1 and Llama-3.1-8B 51.1 (within 2 pp of Sonnet 52.2), suggesting prose-rich layouts tolerate weaker backbones; Mistral-7B-v0.2 lags at 28.6.

ChemTables (drug discovery, n=9). The hardest end of typeset table complexity (image-only and rotated bioactivity tables, footnote-encoded column semantics), but underpowered: the absolute gap is +28.8 F_1 (44.6 best-configuration vs. 15.8 Sonnet few-shot), while the matched per-paper paired bootstrap reports Δ=+8.8 F_1 at p=0.342 (Table 4); only the second is significance-testable. Open 7–8B ArticleMiner runs trail by 12–20 pp F_1 (Mistral-v0.2 19.8, Llama 16.6, Qwen 12.5 vs. Sonnet 32.1) — the 4-tier metric is unavailable on ChemTables, so this is a tuple-F_1 only contrast.

GeoChem (mineral geochemistry, n=28). The ontology-richest and largest-by-row-count domain: 220-entry ontology, 209-column GT schema, 1,236 unique sample IDs, 87,759 non-null element measurements across 73 elements. ArticleMiner reaches 4-tier Overall 76.0 (Sonnet) clustered tightly across closed backbones at 75.5–76.3 (Fig. 4); sample-F_1 is 60.8 [49.9, 76.1] for Sonnet 4.6 with paired $\Delta = +62.3$ F_1 vs. same-LLM PDF few-shot (p<0.001, n=25 common papers; Table 4). The few-shot baseline lands at 0–4 sample-F_1 across all 8 LLMs not because it produces no output (∼3,500 raw `(sample_id, mineral, method, elements)` tuples are emitted across the closed LLMs) but because (i) few-shot uses ad-hoc sample IDs that miss the canonical `sample_name` convention so T_3 structural matching collapses, and (ii) few-shot recovers only rows visible in PDF prose (∼5–30 per paper), missing the hundreds of supplementary-spreadsheet rows that ArticleMiner's deterministic supp-parser stage ingests.

C Ontology Module Entries

Representative entries from the four domain-specific ArticleMiner ontology modules; full modules are released with the source code.

C.1 Geochemistry (220 entries)

The geochemistry ontology has four sub-components. The *deposit-type taxonomy* provides 90 entries spanning the USGS CMiO-MIN hierarchy [1]; each entry is a (*keyword*, *environment*, *group*) triple, e.g. `("mvt", "Basin hydrothermal", "Mississippi Valley-type")`, `("porphyry cu", "Magmatic hydrothermal", "Porphyry")`, `("orogenic gold", "Metamorphic hydrothermal", "Orogenic")`, `("vms", "Volcanic basin hydrothermal", "Volcanogenic massive sulfide")`. The *mineral taxonomy* maps 80 IMA-recognised minerals to material classes, e.g. `("sphalerite", "sulfide", "ZnS")`, `("chalcopyrite", "sulfide", "CuFeS2")`, `("magnetite", "oxide", "Fe3O4")`. The *method standardization* maps 50+ vernacular variants to 12 canonical analytical methods (LA-ICPMS, EPMA, SIMS, ICP-OES, XRF, XANES, ...). The *constraint layer* encodes BDL conventions (-99999), negative-LOD encoding ($< 0.5 \rightarrow -0.5$), N/A as blank, wt%-to-ppm conversion, and per-element plausibility ranges.

C.2 Drug discovery: ChemTables (80 entries)

The bioassay taxonomy enumerates the four standard activity types {IC_{50}, EC_{50}, GI_{50}, MIC} together with their canonical units and dose-response interpretation. The target taxonomy lists representative protein kinases (CHK1, CHK2, EGFR, ...), receptors, and microbial targets seen across the 9 PMC papers, with synonym maps where applicable. Constraints encode dose-response monotonicity and compound-target linkage (each value must be tied to a specific compound identifier).

C.3 Materials science: DiSCoMaT (167 entries)

The material-classification taxonomy distinguishes glass, ceramic, alloy, polymer, plus a long tail of sub-classes from the SciGlass database. The oxide-component taxonomy lists the standard compositional building blocks (SiO_2, Na_2O, B_2O_3, PbO, CaO, ...) with their canonical chemical formulae. Unit standardization distinguishes mol% vs. wt% (both valid; the distinction must be preserved per table). Constraints encode composition-sum tolerance ($\sum \approx 100\%$) and component plausibility ranges per material class.

C.4 Machine learning: MLTables (60 entries)

The entry-type taxonomy enumerates the four MLTables annotation classes {Result, Data Statistic, Hyper-parameter/Architecture, Other}. The metric taxonomy lists common ML metrics (accuracy, F_1, BLEU, perplexity, AUC, mAP, ...) with their value ranges and interpretation (higher-better vs. lower-better). Attribute linking defines the model→dataset→metric→value chain that identifies a Result and the parameter→model association that identifies a Hyper-parameter. Constraints encode metric plausibility (accuracy $\in [0,100]$, loss ≥ 0) and model-dataset consistency within a paper.

D Prompt Templates

This appendix sketches the prompt templates used in each LLM-based stage of ArticleMiner; full templates are released with the source code.

Paper-intelligence Blueprint skim (band 2). System role identifies elements measured, expected sample count, analytical methods, and key domain entities, explicitly forbidding numerical-value extraction (planning step). User input: full PDF text + the variable taxonomy from the ontology module (73 elements with units in geochemistry; four bioassay types in ChemTables; etc.). Output: JSON object `{elements_measured, expected_sample_count, analytical_methods, key_entities}`.

Paper-level metadata LLM (band 2). System role: extract paper-level metadata, validate each category against the provided taxonomy, and reject categories absent from it; never invent units or method names. User input: PDF text, the Blueprint object, and the classification taxonomy (e.g. the 189-class CMMI deposit-type list for geochemistry). Output: JSON conforming to the domain `PaperMetadata` schema with per-category confidence.

Agentic diagnostic LLM (band 3, structural-hint retry). System role: diagnose why a structured-data extractor produced fewer rows than expected, identifying structural problems (wrong header row, transposed layout, off-by-one column offset) with values explicitly forbidden, only structural hints (the iterative refine–feedback pattern of [3] restricted to structural hints). User input: failed-extraction rows, expected sample count, raw table text per backend. Output: JSON `{header_row_index, is_transposed, sample_id_column, units_row_index, notes}`.

Token budget is 4096 for headline runs (8192 for the diagnostic-LLM retry); temperature is 0.0 throughout for reproducibility.

Open-LLM cap. For 7–8B open backbones (Qwen-2.5, Mistral-v0.2, Llama-3.1) on GeoChem, `max_new_tokens` is capped at 512 across blueprint, metadata, table-extraction, and deposit-classifier stages. The cap suppresses hallucination spirals observed under longer budgets (e.g., open-ended isotope-chain enumeration on geochemistry tables) and short-circuits failed JSON-schema completions; the deposit classifier still emits malformed JSON on the 189-class CMMI taxonomy under all three open backbones, which the pipeline catches and writes `deposit_type=null` (§4.2, "Open-LLM geochemistry").

Few-shot baseline prompt (matched-input comparison). The same-LLM PDF few-shot baseline used throughout Tables 1, 4 (and the equivalents at the pre-parsed input modality can also be found in Table 1) is implemented as a single-call ICL prompt at temperature 0 with a fixed three-exemplar prefix. The exemplars are drawn from the train split of each benchmark (DiSCoMaT, ChemTables, MLTables) or, for GeoChem, from a held-out exemplar pool not used in evaluation. The system role is: *"Given the paper text below, return a JSON list of `<tuple-shape>` records. Output JSON only."* `<tuple-shape>` is the per-domain tuple shape from §A. The user message is exactly the same input given to ArticleMiner's extraction stage (full PDF text or pre-parsed table text per modality), without any blueprint, ontology context, vision fallback, or self-correction. Exemplars are formatted identically: each is a `<input-block>`→`<JSON-list>` pair. We re-use the same exemplars across all eight LLM backbones, so the only thing that varies between rows in Table 1 is the LLM. Token budget is 4096 input / 8192 output (the same as ArticleMiner's extraction stage); on GeoChem, this is sufficient to emit the few-shot's per-row tuples for all 28 papers (the few-shot baseline does not attempt to match the 209-column GT schema, hence the 0–4 sample-F_1 band reported in Table 1 despite ~3,500 raw rows being emitted across the closed LLMs). Three-exemplar count was chosen on ChemTables (k-shot scaling, Fig. 5): the baseline F_1 plateaus at $k = 3$, so additional exemplars are not used.

The released code in `src/runners/run_fewshot_pdf.py` contains the verbatim prompt strings, exemplar pool, and per-domain tuple shapes.

E Sonnet 4.6 Sensitivity on PDF-only Domains

The main-text Tables 2 and 3 fix the LLM backbone to Haiku 4.5 for cross-domain uniformity. On the small PDF-only benchmarks ChemTables (n=9) and MLTables (n=15), Haiku's smaller absolute F_1 floor compresses ablation deltas to within 4–10 pp. Re-running the same ablations on Sonnet 4.6 reveals component effects an order of magnitude larger, providing a sharper view of which stages drive the architecture's value when the backbone has more headroom.

Table 1: **Sonnet 4.6 component ablation** on ChemTables/MLTables, companion to Table 2. Vision dominates on PDF-only domains (−17.5 / −47.4 F_1 when removed); removing paper-intelligence *boosts* ChemTables by +12.5 F_1 via stricter precision. Bold = largest drop / largest gain.

	ChemTables (n=9)			**MLTables** (n=15)		
Variant	P	R	F_1	P	R	F_1
Full pipeline	30.4	34.0	32.1	52.6	51.7	52.2
w/o ontology	28.8	32.3	30.4	51.9	51.6	51.7
w/o self-correction	28.6	32.3	30.3	51.8	51.6	51.7
w/o validation	28.4	32.3	30.2	51.7	50.8	51.2
w/o intelligence	**64.9**	34.0	**44.6**$^{\uparrow}$	52.5	52.1	52.3
w/o vision	17.1	12.8	**14.6**	28.3	2.6	**4.8**

$^{\uparrow}$ removing the paper-intelligence stage raises ChemTables F_1 by +12.5 — the over-extraction filter is over-cautious on bioassay tables. Cross-model takeaway: weaker backbones (Haiku) attenuate ablation effects everywhere, while stronger backbones (Sonnet) amplify them where benchmark size allows the gap to register; both share the consistent finding that GeoChem's supplementary-data path makes LLM-side components nearly inert.

F Extended Results

F.1 GeoChem per-paper 4-tier breakdown ($n = 28$)

Per-paper $T_1/T_2/T_3/T_4$ scores for the headline configuration (**Full (5-backend, ArticleMiner)** from Table 9, Sonnet 4.6) yield the following bootstrap 95% CIs on per-tier means (n=28, 1,000 resamples, seed 42): T_1 (metadata) 71.6 [68.6,74.7], T_2 (numerical) 76.5 [65.4,87.2], T_3 (structural) 68.5 [55.0,80.4], T_4 (null) 90.6 [85.2,95.1], **Overall** 76.0 [70.1,81.4]. The T_2 and T_3 intervals are wide (±10–13) because per-paper scores have a long left tail (a small number of layout-only or borderless-table papers score near 0); the overall-mean interval (±5.7) is correspondingly conservative.

F.2 Cross-domain ablation referrals.

The component ablations (Table 2) and PDF-backend ablations (Table 3) in the main text cover all four domains uniformly under Haiku 4.5. Three observations are worth restating in the supplementary:

(1) Ontology weight scales with domain richness. Removing the ontology costs -16.3 F_1 on DiSCoMaT (oxide-component validation is the dominant guard) and -7.7 on MLTables, but only -0.7 on GeoChem (supplementary-spreadsheet evidence already encodes most ontology constraints) and -2.2 on the small ChemTables benchmark.

(2) Two ablations *improve* over the full pipeline. Removing intelligence gains +4.1 on DiSCoMaT; removing vision gains +1.1 on GeoChem. Both signal noise sources where the heuristic over-extracts, captured in Table 2 with the $\uparrow$ marker.

(3) Backend dominance is per-domain. On DiSCoMaT, single `docling` reaches 71.7 F_1 and removing it collapses to 43.5; the other four backends barely contribute, so consensus is essentially neutral. On ChemTables/MLTables, single `docling` (26.2 / 56.3) beats the full 5-backend pipeline (20.8 / 47.6) — multi-backend over-extracts noise on small benchmarks. On GeoChem, all single/loo variants cluster tightly at 46–48 F_1 versus the full pipeline's 54.3 — the supplementary-spreadsheet path dominates row content. `Camelot` is the most consistently harmful backend cross-domain: removing it improves F_1 on ChemTables (+4.0), MLTables (+7.7), and DiSCoMaT (+1.5).

F.3 Open-LLM GeoChem 4-tier breakdown ($n = 26$, max-tokens=512)

The three open 7–8B backbones run the full ArticleMiner pipeline on the 26-paper subset (the 2 data-reuse entries lacking standalone PDFs are excluded; see annotation protocol). Per-LLM 4-tier and sample-F_1:

- Qwen-2.5-7B: T_1=22.2, T_2=64.3, T_3=60.2, T_4=83.5, Overall = 54.0; sample F_1=57.4.
- Mistral-7B-v0.2: T_1=17.7, T_2=64.3, T_3=60.2, T_4=83.5, Overall = 52.6; sample F_1=57.4.
- Llama-3.1-8B: T_1=20.5, T_2=64.3, T_3=60.2, T_4=83.5, Overall = 53.4; sample F_1=57.4.

$T_2/T_3/T_4$ are bit-identical across the three backbones because the deterministic supplementary-spreadsheet parser dominates row count and numeric content on GeoChem; the T_1 spread (4.5 pp) tracks per-LLM metadata-extraction quality after the deposit classifier returns null on all three (malformed JSON on the 189-class CMMI taxonomy). Per-paper inspection (`Chu_et_al_2022.pdf`) confirms identical row counts (160) and identical `mineral`/`deposit_name`/`analytical_method` extractions across the three open LLMs but with cell-level differences in element values and ancillary fields.

G Hardware, Software, and Reproducibility

All experiments ran on a single Linux workstation (Ubuntu 22.04, CUDA 12.4) with NVIDIA H100 80GB GPUs for open-weight LLM inference (vLLM 0.6.x); closed-source backbones were called via the official Anthropic, OpenAI, and Google Gemini SDKs (Apr–May 2026). Pinned model identifiers: `claude-sonnet-4-6`, `claude-opus-4-6`, `claude-haiku-4-5-20251001`, `gpt-4o-2024-08-06`, `gemini-2.5-flash`, `Qwen-2.5-7B-Instruct`, `Mistral-7B-Instruct-v0.2`, `Llama-3.1-8B-Instruct`. PDF backends: `docling 2.x`, `marker 1.x`, `mineru 0.9` (PaddleOCR weights unavailable in our sandbox; mineru is monkey-patched to no-op when its OCR path triggers, so it contributes table-detection metadata only), `pdfplumber 0.11`, `camelot-py 0.11`. Evaluator code and the four ontology modules are released under CC-BY-4.0 at https://anonymous.4open.science/r/articleminer-iswc26. Seed 42 throughout (paired bootstrap, sample shuffling, exemplar order). All API calls use temperature 0; `max_tokens` caps are documented per stage in the released config. Cost/runtime per paper is reported in Table 7; the cross-domain ablation grid (5 component variants × 11 backend variants × 4 domains) costs ∼\$25 on Haiku 4.5 and would be ∼\$80 on Sonnet 4.6 — the choice of Haiku 4.5 as the ablation backbone makes the full grid affordable on a 163-paper corpus.

H Knowledge-Graph Output: JSON-LD Example

This section shows one complete GeoChem `Sample` record as it appears in ArticleMiner's emitted KG. The record is derived from the `Chu_et_al_2022` paper (sphalerite from a basin-hydrothermal Pb-Zn deposit). The same record is also serialized to Turtle; both serializations share a single source of truth in the IRI declarations of the geochemistry ontology module Ω_{geochem} (§C).

```
{
  "@context": {
    "ima":  "http://rruff.info/ima/",
    "cmio": "https://anonymous.example/cmio-min/",
    "am":   "https://anonymous.example/articleminer/",
    "prov": "http://www.w3.org/ns/prov#",
    "value_ppm": {"@id": "am:valuePpm",  "@type": "xsd:decimal"},
    "lod_ppm":   {"@id": "am:lodPpm",    "@type": "xsd:decimal"}
  },
  "@id":      "am:sample/Chu2022/LGB-Sp1-spot7",
  "@type":    "am:Sample",
  "sample_name":       "LGB Sp1 spot7",
  "sample_local_id":   "LGB-Sp1-spot7",
  "analysis_id":       "LA-ICP-MS-LGB-Sp1-spot7",
  "mineral":          {"@id": "ima:sphalerite"},
  "analytical_method": "LA-ICPMS",
  "deposit_name":      "Lago Verde basin",
  "deposit_type":      {"@id": "cmio:mvt"},
  "deposit_environment":"Basin hydrothermal",
  "deposit_group":     "Mississippi Valley-type",
  "country":          "Italy",
  "publication_date":  "2022",
  "measurements": [
    {"element": "Fe", "value_ppm": 22929,    "lod_ppm": 1.2},
    {"element": "Cu", "value_ppm": 1507,     "lod_ppm": 0.4},
    {"element": "Cd", "value_ppm": 1846,     "lod_ppm": 0.3},
    {"element": "In", "value_ppm": -99999,   "lod_ppm": 0.05},
    {"element": "Mn", "value_ppm": null,     "lod_ppm": null}
  ],
  "prov:wasDerivedFrom": "doi:10.xxxx/chu2022",
  "prov:wasGeneratedBy": {
    "@type": "am:ExtractionRun",
    "am:backendSupport":   0.83,
    "am:sourceBackend":    "docling",
    "am:sourceLocator":    "Chu2022.pdf#page=4-table-2"
  }
}
```

Listing 1.1: JSON-LD serialisation of a single GeoChem sample (excerpt). The `ima:` CURIE links `mineral` to the IMA list [4]; `cmio:` CURIEs link `deposit_type` to the CMiO-MIN hierarchy [1]; `prov:` predicates record extraction support and source. The `value_ppm: -99999` sentinel encodes below-detection-limit under a closed-world assertion.

Equivalent Turtle (excerpt).

```
@prefix am:   <https://anonymous.example/articleminer/> .
@prefix ima:  <http://rruff.info/ima/> .
@prefix cmio: <https://anonymous.example/cmio-min/> .
@prefix prov: <http://www.w3.org/ns/prov#> .

am:sample/Chu2022/LGB-Sp1-spot7 a am:Sample ;
    am:sampleName       "LGB Sp1 spot7" ;
    am:mineral         ima:sphalerite ;
    am:analyticalMethod "LA-ICPMS" ;
    am:depositType      cmio:mvt ;
    am:hasMeasurement [ am:element "Fe" ; am:valuePpm 22929 ; am:lodPpm 1.2 ] ,
                      [ am:element "In" ; am:valuePpm -99999 ; am:lodPpm 0.05 ] ;
    prov:wasGeneratedBy [ a am:ExtractionRun ;
                          am:backendSupport 0.83 ;
                          am:sourceBackend "docling" ] .
```

Downstream queries. The above record is queryable via SPARQL alongside MinMod [2] `MineralSite` records. Example: *"all chalcopyrite samples with detected Cu"*:

```
SELECT ?s ?cu WHERE {
  ?s a am:Sample ;
     am:mineral ima:chalcopyrite ;
     am:hasMeasurement [ am:element "Cu" ; am:valuePpm ?cu ] .
  FILTER(?cu > 0)         # excludes BDL sentinel -99999
}
```

ArticleMiner's GeoChem run for this paper emits 28 `MineralSite` records (one per paper) directly ingestible at the MinMod SPARQL endpoint; ChemTables/MLTables/DiSCoMaT runs emit JSON-LD only (no MinMod CDR JSON, since those domains do not have a deposit-graph target). The constraint layer Γ_d described in §2.1 is expressible as SHACL shapes over these classes (an admissibility predicate maps to `sh:in`, a numeric range to `sh:minInclusive`/`sh:maxInclusive`, class membership to `sh:class`); the same validators that gate emission inside the pipeline can therefore be applied externally by SHACL-aware tooling.

I GeoChem-28 Annotation Protocol

This section summarises the curation protocol used for the 28 contributed mineral-geochemistry papers. The full protocol document is released alongside the dataset at `data/geochem28/annotation_protocol.md`.

Scope. 28 peer-reviewed papers reporting LA-ICP-MS trace-element analyses of sulfide and sulphosalt minerals in ore deposits, published 2004–2025. Each paper is annotated against the 209-column CMiO-MIN schema (an extension of CMiO/Hofstra [1] for individual-grain mineral geochemistry).

Annotators. Multiple geochemistry annotators participated, coordinated with U.S. Geological Survey domain experts. Annotations were independently spot-checked by a second annotator and any disagreements reconciled against the source PDF + supplementary tables.

Per-paper procedure.

1. **Deposit metadata (T_1).** Fifteen deposit and analytical method fields are extracted (`deposit_name`, `deposit_type`, `deposit_environment`, `deposit_group`, `all_commodities`, `mineral`, `analytical_method`, `instrument_type_model`, `laboratory_location`, `operating_conditions`, `standards_used`, `country`, `age`, `host_rock`, `sample_reference`). Values are taken verbatim from the paper text when possible, normalized to the controlled vocabulary in Ω_{geochem} otherwise.
2. **Per-sample rows ($T_2/T_3/T_4$).** Each row corresponds to one `(sample_id, mineral)` analysis under a three-tier sample ID (top-level sample / sub-sample or thin section/spot). For each element column `<el>_ppm` or `<el>_wtpct`: a numeric value if reported; -99999 (five nines, negative) if explicitly reported as below detection limit; blank if not measured / not reported.

3. **Mineral assignment.** One mineral per row. Mineral identity is taken from the per-analysis annotation in the paper's supp tables (`analysis_id` prefix, data-sheet name, or explicit column), never inferred from abundance patterns.
4. **Units.** Preserved as reported; no conversion. Columns carry the unit in their name (e.g. `cu_ppm`, `s_wtpct`).

Conventions.

– **Below-detection limit:** −99999 (five nines, negative). This sentinel distinguishes BDL from *not measured* (blank), enabling the T_4 null-pattern dimension that no prior table-IE benchmark scores.
– **Not applicable / not measured:** blank cell.
– **Reference materials** (MASS-1, NIST 610, etc.) are included as sample rows tagged with the reference-material flag. Annotation does not filter them out; both ArticleMiner and the few-shot baseline are expected to reproduce them in the prediction set so that the structural match in T_3 is well-defined.
– **Data-reuse papers** (suffix `_as_reported_in_X_et_al_YYYY`): gold tuples are taken from paper X's supplementary tables. The original paper PDF is therefore not strictly required for evaluation, because all gold numbers reappear in the companion paper. Two of the 28 entries are data-reuse pairs; the open-LLM 4-tier breakdown in §F.3 reports $n = 26$ rather than $n = 28$ because the open-LLM PDF-vision path requires standalone PDFs (not available for the two reuse entries), whereas the closed-LLM and headline runs report $n = 28$ because they ingest the supplementary spreadsheets directly.

Licence. Ground-truth annotations are released under CC-BY-4.0. Source-paper PDFs remain under publishers' original licenses and are included in the benchmark release under fair use for academic reproducibility.

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